

4(a)(i)	(caused by) <u>pathogen</u> and , transmissable / AW ;	1
4(a)(ii)	any three from: 1 ref. to presence of, (B / T) memory cells from primary immune response ; 2 increased chance of memory cells encountering antigen ; 3 more lymphocytes, with receptors complementary / specific, to antigens ; 4 more plasma cells created ; 5 increased / faster, production of antibodies ;	3
4(b)(i)	passive and artificial ;	1
4(b)(ii)	any two from: - temporary / short term ; - immediate effect ; - involves injection of, antibodies immunoglobulin ; - does not, stimulate the immune system / initiate a primary immune response ; - AVP ; e.g. no memory cells produced - no new antibodies made	2
4(c)(i)	<i>Mycobacterium tuberculosis</i> / <i>Mycobacterium bovis</i> ;	1

Question	Answer	Marks
4(c)(ii)	any four from: <i>treatment (max 3):</i> 1 combining rifampicin treatment with drugs, that keep rifampicin inside the bacteria / indigestion drugs ; 2 rifampicin will kill bacteria (more, effectively / quickly) ; 3 reduce length of time needed for successful treatment of TB (from 6 months) ; 4 shorter treatment increases likelihood that treatment will be completed ; 5 idea that use of, widespread / common, indigestion drugs should reduce costs of (usually) long-term TB treatment ; 6 less need for DOTS for some people ; <i>reduce chance of resistance:</i> 7 (if the length of treatment is shorter there is) less time for a mutation to occur in bacterial genome resulting in resistance to rifampicin ; 8 AVP ; e.g. dea that the indigestion drugs may be inhibiting membrane pumps in the cell surface membranes of TB bacteria	4

Question	Answer	Marks
----------	--------	-------